

Gradient Descent Manual Calculation.

1. Computing the predicted values $\hat{y}$ for each data point

$$\text{Model: } \hat{y} = (m_1 \times x_1) + (m_2 \times x_2) + b$$

x_1 and x_2 are input features
 m_1, m_2 and b are parameters.

For Sample 1: $x_1 = 1, x_2 = 3$ and $y = 5$
Sample 2: $x_1 = 4, x_2 = 10$ and $y = 6$
 $m_1 = -1, m_2 = 2.$

$$\begin{aligned}\hat{y}_1 &= (m_1 \times x_1) + (m_2 \times x_2) + b \\ &= (-1 \times 1) + (2 \times 3) + 1 \\ &= (-1 \times 6) + 1 \\ &= -6 + 1 \\ \hat{y}_1 &= -5\end{aligned}$$

$$\begin{aligned}\hat{y}_2 &= (-1 \times 4) + (2 \times 10) + 1 \\ &= (-4 \times 20) + 1 \\ &= -80 + 1 \\ \hat{y}_2 &= -79\end{aligned}$$

$$\begin{aligned}e_1 &= \hat{y}_1 - y_1 \\ &= -5 - 5 \\ e_1 &= -10\end{aligned}$$

$$\begin{aligned}e_2 &= \hat{y}_2 - y_2 \\ &= -79 - 6 \\ e_2 &= -85\end{aligned}$$

$$\therefore \underline{\underline{e = [-10, -85]}}$$